

Problem-Driven Growth Theory: Knowledge Creation at the Frontier

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Abstract

This paper proposes the Problem-Driven Growth Theory. The central claim is that growth at the frontier is generated by the interaction of **problem pressure** and **problem-solving capacity**. Problem pressure captures the gap between the actual and the attainable performance of an economy. Capacity captures the ability of a society to search, experiment, and generate new explanatory knowledge. When both are high, knowledge creation accelerates, technological change occurs, and growth persists. When either is low, stagnation or collapse follows. This mechanism is formalised in a reduced-form model, linked to existing theories in economics, and illustrated through historical cases including Japan, Silicon Valley, Ming China, Venice, Manchester, and Merv. The theory yields falsifiable predictions, offers an organising framework for diverse literatures on growth and decline, and suggests policy implications for innovation and resilience. The novelty of the theory is to treat knowledge creation as endogenous to the interaction of pressure and capacity.

1 Introduction

Economic growth at the frontier is uneven. Some societies grow rapidly, some stagnate for decades, and some collapse. Manchester during the industrial revolution, Silicon Valley in the late twentieth century, and medieval Venice all illustrate periods of intense knowledge creation. By contrast, post-1990 Japan displays decades of stagnation despite being highly educated and technologically advanced (Japan's GDP growth since 1990 0.8% per year). Merv, once a major Silk Road city, vanished after violent destruction (Merv destruction dates: 1221–1223). These patterns motivate a simple question: *why do new ideas appear when and where they do?*

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Modern growth theory has offered several answers. Endogenous growth theory formalised the role of knowledge as a non-rival input that does not face diminishing returns. In this view, ideas can be used simultaneously by many agents and therefore drive long-run growth. Evolutionary theories emphasise variation, selection, and cumulative change. Institutional theories argue that political and legal systems shape incentives for innovation. Historical work has traced how technologies, trade, and institutions co-evolve. Yet across these literatures, the origin of new knowledge is frequently treated as exogenous, automatic, or institutionally determined.

This paper takes a different perspective. The central mechanism proposed here is that knowledge creation is driven by **problems**. Societies face constraints, mismatches between current performance and what could be achieved given existing or potential knowledge. These gaps generate dissatisfaction, incentives to experiment, and in some cases existential urgency. When a society possesses both **problem pressure** and **problem-solving capacity**, new explanatory knowledge is produced. When either dimension is absent, growth stagnates or collapses. I refer to this mechanism as the **Problem-Driven Growth Theory**.

The idea is intuitive. Problems provide the demand for knowledge; capacity provides the supply. When problems are large and capacity is strong, search and experimentation occur. When problems are large but capacity is weak, collapse may follow. When capacity is high but problems are mild, societies may enjoy stability but little innovation. When both are low, decay occurs. These regimes generate clear predictions for growth at the frontier.

Historical cases illustrate the mechanism. Japan's stagnation after 1990 is consistent with moderate problem pressure and high capacity: without major shocks, innovation slows. Silicon Valley's rise reflects high pressure (military, scientific, commercial) and high capacity (universities, venture capital, legal systems, dense networks). Ming China's decline reflects the reduction of external pressures and constraints on merchants. Venice declined after institutional closure reduced variation and barriers to entry increased. Manchester's industrial explosion reflects severe energy and labour constraints combined with coal, transport, and legal innovation. Merv collapsed when violent destruction eliminated capacity.

These examples motivate a reduced-form model of frontier growth. The next section situates the theory within existing literature.

2 Literature Background

This section situates the Problem-Driven Growth Theory within existing work. The aim is not to review all literature on growth, but to clarify how current theories explain long-run development and where gaps remain. The central question motivating this paper is why new knowledge is produced in some places and periods, but not in others. All theories reviewed

here contain partial answers, and the mechanism proposed in this paper complements them by focusing on the role of problems as drivers of knowledge creation.

2.1 Endogenous Growth and Non-Rival Ideas

Endogenous growth theory established that long-run growth is driven by the accumulation of knowledge. Romer (1990) showed that ideas are non-rival inputs: a piece of knowledge can be used simultaneously by many agents without diminishing its usefulness. This property implies that the standard diminishing returns logic of physical capital does not apply. Investments in research, education, and technological capabilities can therefore produce sustained increases in output. In Romer’s model, research effort generates new ideas, which enter the production function and raise productivity.

This framework explains how rich countries can grow indefinitely. However, it assumes that knowledge creation is a function of research effort without specifying when or why research effort intensifies. In periods of stagnation, research effort may remain high, yet technological change slows. Conversely, in periods of intense pressure—such as wartime, energy crises, or emerging industries—search is more aggressive, and breakthroughs appear. The model proposed in this paper complements Romer’s approach by identifying *problem pressure* as a mechanism that induces search.

2.2 Evolutionary Theories of Innovation

Evolutionary models treat innovation as a process of variation and selection. Nelson and Winter (1982) describe firms as having routines rather than perfect rationality. Firms experiment, imitate, and adapt. Successful behaviours are retained through selection. This perspective captures cumulative change, path dependence, and the importance of diversity.

However, evolutionary models typically treat variation as exogenous or constant. They do not explain the conditions under which variation intensifies. The Problem-Driven Growth Theory suggests that problems influence the intensity of variation. When gaps between actual and attainable performance grow, societies explore more aggressively. Variation increases. This is consistent with historical episodes in which crises triggered bursts of innovation: codebreaking during war, atomic research, vaccine development, or energy exploration.

2.3 Institutions and Incentives

Institutional theories emphasise incentives for innovation. Acemoglu and Robinson (2012) argue that inclusive institutions allow broad participation and protect property rights, while extractive institutions limit entry and concentrate power. This framework explains why some

societies discourage innovation: powerful groups may block it to preserve rents, as in Venice after the Serrata. Institutions shape the incentives to create and adopt new knowledge.

However, institutions alone cannot account for the timing of innovation. Some inclusive societies innovate little; some extractive societies innovate when under threat. Institutional incentives matter, but they are not sufficient. The Problem-Driven Growth Theory emphasises that institutions are part of *capacity*, while problems are part of *pressure*. Only when both align does knowledge creation accelerate.

2.4 Induced Innovation

Hicks proposed that changes in relative prices induce innovation. When input prices increase, firms search for substitutes. This view has been used to explain energy-saving technologies and labour-saving machinery. Induced innovation supports the idea that dissatisfaction or shortage triggers search. However, price changes represent only one form of problem. Structural constraints, existential threats, or strategic competition may generate problems even when prices remain stable. The broader concept of problem pressure therefore extends induced innovation.

2.5 Knowledge and Useful Knowledge

Mokyr distinguishes between propositional knowledge (knowledge of how the world works) and prescriptive knowledge (how to do something). The Industrial Revolution, in this view, was driven by a cultural and institutional environment that promoted the accumulation and dissemination of useful knowledge. Artisans, scientists, and entrepreneurs interacted in networks, exchanging ideas and solving problems. Knowledge flowed between domains, and practical applications emerged.

This historical account highlights the importance of explanation. Problems were solved not only by trial-and-error but by understanding underlying mechanisms. The Problem-Driven Growth Theory complements this by emphasising the conditions under which societies search for explanation. When pressure is high and capacity exists, useful knowledge expands.

2.6 Frontier Growth and Stagnation

Theories of convergence suggest that poorer countries grow faster by adopting existing technologies. Yet even among rich countries, growth diverges. The frontier moves through the creation of new technologies, not only through adoption. Some frontier economies stagnate, others accelerate. Existing models treat this divergence as a function of institutions, culture, or accumulation. The Problem-Driven Growth Theory argues that frontier divergence

reflects the intensity of problems and the ability to respond.

Japan after 1990 provides an example: high education, strong institutions, and capital accumulation coexisted with slow growth. Without urgent pressure or existential challenges, innovation slowed. Conversely, Silicon Valley maintained pressure from rapid technological change, military competition, and scaling challenges, combined with capacity from universities, immigration, and capital. Frontier growth persisted.

2.7 Collapse and Irreversibility

Economic collapse is often underexplained. Civilisations such as Merv or post-Roman Britain did not merely stagnate; they lost capabilities. The destruction of capacity—population, institutions, knowledge, or infrastructure—reduces the ability to solve problems. Even if pressure remains high, collapse may follow. The Problem-Driven Growth Theory emphasises that resilience matters: capacity must survive shocks. This observation is consistent with archaeological and historical accounts of irreversible collapse.

2.8 Relation to the Present Theory

Existing theories capture important aspects of growth:

- Endogenous growth highlights the role of ideas.
- Evolutionary models highlight variation and selection.
- Institutional theories highlight incentives.
- Induced innovation highlights search.
- Historical work highlights explanation and useful knowledge.

The mechanism proposed here does not replace these views. It provides a unifying structure. Problems generate demand for knowledge, while capacity determines supply. The interaction of pressure and capacity explains why growth accelerates, stagnates, or collapses. The next section formalises the concepts of problem pressure and capacity.

3 Problem Pressure

This section defines problem pressure, explains why it matters for knowledge creation, and considers how it may be measured. The core intuition is straightforward. Societies do not innovate unless they have something to solve. A problem, in this context, is a gap between

current performance and attainable performance given existing or potential knowledge. The greater the gap, the stronger the incentive to search, experiment, and adopt new solutions.

3.1 Definition

Let Y_t denote actual economic capability at time t : what a society is currently able to produce given its knowledge, technology and organisation. Let $\bar{Y}_t$ denote the attainable frontier at time t : what could be produced in principle if current explanatory knowledge were fully and optimally applied.

It is important to distinguish between the existence of problems and the realisation of problems. Human societies have faced constraints for hundreds of thousands of years. Hunger, disease, scarcity and conflict were persistent. Yet for most of history there was almost no sustained growth. This suggests that problem pressure is not driven by objective conditions alone. A problem must be recognised as a problem and believed to be solvable.

We therefore define:

$$P_t = \phi_t(\bar{Y}_t - Y_t)$$

where $\phi_t \in [0, 1]$ measures the share of the gap that is perceived, salient and actionable. For most of human history $\phi_t \approx 0$: problems were interpreted as fate or the will of gods, and not as opportunities for explanation or improvement. Only when societies develop the epistemic norm that problems are solvable does realised problem pressure increase. As explanatory knowledge grows, new problems continuously emerge at higher levels of capability.

3.2 Economic Interpretation

Problem pressure measures dissatisfaction with the current state of affairs. This may arise from:

- stagnant productivity,
- unmet demand for goods or services,
- resource shortages,
- geopolitical threats,
- technological competition,
- demographic pressures,
- environmental constraints.

The concept is deliberately broad. Different societies experience different pressures. What matters is not the source of the problem but its existence. When a society recognises a gap between what is and what could be, incentives to search appear. This search is not necessarily optimising in a strict sense; it may involve trial-and-error, learning, imitation, or experimentation.

3.3 Types of Problem Pressure

Problem pressure can be classified into several types. These categories are not exhaustive, but they illustrate the range of forces that induce knowledge creation.

3.3.1 Economic Pressure

Economic pressure arises when demand exceeds supply, or when production becomes inefficient. Historically, constraints on energy, labour, or transport have generated innovation. Manchester during the Industrial Revolution faced high energy costs and labour constraints. The search for waterpower, coal, and machinery was driven by necessity. Economic pressure induces technologies that economise on scarce inputs or create new substitutes.

A classic illustration of this mechanism is due to Hicks (1932). He argued that when the price of a factor rises, firms face a problem: production becomes more costly. The higher price reveals scarcity and increases the gap between current and attainable performance. This pressure stimulates invention, leading to labour-saving or energy-saving technologies. In this view, price signals are one channel through which problem pressure induces search and innovation.

3.3.2 Technological Pressure

Technological pressure arises when existing solutions fail to scale or meet new requirements. Silicon Valley provides an example. As computing demands increased, problems emerged in storage, speed, networking, and cryptography. These problems generated incentives to search. New knowledge was created because new problems appeared continually.

3.3.3 Geopolitical and Military Pressure

Strategic threats have repeatedly triggered bursts of innovation. The development of radar, cryptography, aircraft, and nuclear technology were all driven by wartime pressures. Military urgency compresses time horizons and prioritises problem-solving. The scale of resources mobilised to solve these problems reflects the magnitude of pressure. After the threat passes, pressure may decline, and innovation may slow.

3.3.4 Social and Demographic Pressure

Population growth, ageing, disease, and migration create constraints. Public health crises generate urgency. The development of vaccines and medical technologies is often problem-driven. Demographic pressures induce institutional reforms, changes in education, and social innovations. These are forms of knowledge that shape economic outcomes.

3.4 Frontier Economies and Pressure

Frontier economies differ from developing economies in a fundamental way. For developing economies, growth can occur through adoption of existing technologies. For frontier economies, growth requires the creation of new knowledge. This implies that frontier economies must continuously generate new problems in order to sustain growth. When pressures are mild, stagnation may occur. When pressures are severe but capacity is high, growth accelerates.

This dynamic can explain why peaceful, stable, wealthy societies often stagnate. The absence of urgency reduces search intensity. Incremental improvements occur, but the gap P_t remains small. In contrast, societies experiencing strong competitive pressures or existential threats may innovate rapidly.

3.5 Measurement

Problem pressure is not directly observable, but several proxies are possible. These include:

- productivity gaps relative to best-practice firms or countries,
- shortages in inputs such as labour, energy, or raw materials,
- price signals, such as spikes in wages or resource costs,
- defence spending as a share of output,
- unmet demand in housing, healthcare, or education,
- demographic indicators such as dependency ratios,
- rate of obsolescence in technologies.

None of these measures alone captures the full concept, but together they approximate the intensity of pressure. This also implies that problem pressure is multidimensional. Societies may face economic pressure without military pressure, or vice versa. What matters for knowledge creation is the overall magnitude.

3.6 Dynamism of Problem Pressure

Problem pressure is dynamic. As new knowledge and technologies improve performance, the frontier $\bar{Y}_t$ moves upward. Even if current problems are solved, new ones appear. This mechanism explains why growth at the frontier can be unbounded. Solving problems expands capability, which creates new problems at a higher level of complexity. This relationship is central to understanding long-run growth.

3.7 Illustrative Cases

Historical cases illustrate the role of pressure. Manchester faced severe constraints on energy and labour, leading to innovations in machinery and coal-based power. Silicon Valley faced scaling problems in computing, leading to semiconductor and software breakthroughs. By contrast, post-1990 Japan faced moderate pressure, leading to incremental improvements rather than transformative innovation. Merv faced high pressure but lost capacity through destruction, resulting in collapse. Venice faced declining pressure as trade patterns changed and institutions closed.

These examples support the claim that problem pressure is necessary but not sufficient. Without capacity, pressure leads to collapse. Without pressure, capacity leads to stagnation. Only the combination leads to knowledge creation.

The next section discusses the second component of the theory: problem-solving capacity.

4 Problem-Solving Capacity

Problem pressure is necessary for growth, but not sufficient. Societies may face severe constraints without being able to respond. The second component of the theory is **problem-solving capacity**, or the ability of a society to search, experiment, generate, and adopt new explanatory knowledge. This idea is consistent with a long tradition in economics and philosophy: innovation requires both the presence of problems and the ability to solve them.

4.1 Definition

Define problem-solving capacity as a function of human capital, institutions, material resources, and resilience:

$$C_t = f(H_t, I_t, M_t, R_t) \tag{1}$$

Capacity is high when societies allow entry, encourage experimentation, mobilise resources, and survive failures.

4.2 Human Capital and Diffusion

The idea that human capital enables the diffusion of ideas has deep roots. Nelson and Phelps (1966) argued that education increases the speed at which countries adopt existing technologies. Benhabib and Spiegel (1994) showed empirically that countries with higher levels of human capital converge faster to the frontier. Barro and Sala-i-Martin (1997) demonstrated that growth rates depend strongly on schooling and skills. These results support the idea that human capital increases the ability to recognise problems and implement solutions.

Education does more than transmit existing techniques. Scientific training enables the creation of *explanatory knowledge*: general theories that apply across contexts (Deutsch, 2011). Explanatory knowledge is deeper and more powerful than mere rule-of-thumb procedures. It allows societies to solve unfamiliar problems, not only familiar ones. Human capital therefore contributes directly to the creation of new knowledge, not merely its adoption.

4.3 Institutions and Error-Correction

Institutions determine who may experiment and how errors are handled. Popper (1959) argued that knowledge grows through a process of conjectures and refutations: progress occurs when errors are eliminated. This insight connects directly to innovation. Societies must be able to tolerate failure, criticism, and revision. Inclusive institutions protect the right to experiment and the right to dissent. Extractive institutions suppress variation and centralise authority, reducing the scope for error correction.

Venice illustrates this dynamic. During its ascent, Venice allowed merchants to engage in trade, shipbuilding, and finance. Variation was abundant, and successful practices spread. After the Serrata, political participation was restricted, and families closed entry. Variation declined. The ability to correct errors diminished. Even as pressures remained, capacity fell.

4.4 Material Resources and Infrastructure

Material resources include finance, infrastructure, and physical capital. Laboratories, factories, roads, ports, and energy systems enable search and implementation. Finance allows experimentation under uncertainty. Infrastructure supports diffusion. However, resources are not sufficient. Societies with abundant resources may stagnate if institutions restrict entry or if resilience is weak. Capacity emerges from the interaction of components, not from any single factor.

4.5 Resilience and Survival of Error

Resilience is essential for innovation. Error is unavoidable. Most experiments fail. Popper's logic of error-correction implies that progress requires the ability to survive mistakes. Deutsch (2011) emphasises that explanatory knowledge is created when errors are found and eliminated. Societies that cannot survive error cannot correct it.

Resilience may arise from:

- redundancy in systems,
- decentralised decision-making,
- social cohesion and trust,
- reserves of capital or food,
- military capability.

Historical collapses illustrate the importance of resilience. Merv possessed knowledge, infrastructure, and trade networks, but lacked resilience. When destroyed, knowledge and institutions were lost. Problems remained, but capacity disappeared.

4.6 Substitutability and Complementarity

Capacity is multidimensional. Human capital may substitute for resources; institutions may substitute for infrastructure. But some thresholds cannot be crossed by substitution. Education cannot substitute for physical survival. If collapse eliminates population, knowledge is lost. Institutions cannot substitute for food in a famine. Complementarities matter: each component enables others.

4.7 Fragility of Capacity

Capacity is fragile. It can be destroyed quickly. Societies that take generations to build institutions, knowledge, and trust may lose them in a short period. Collapse is easier than creation. This asymmetry is consistent with Popper's insight: error correction depends on survival. If errors lead to destruction rather than correction, knowledge cannot grow.

4.8 Link to Frontier Growth

Frontier economies require constant problem-solving. As the attainable frontier moves upward, new problems appear. Societies must respond. Capacity determines whether they can. When problem pressure and capacity align, explanatory knowledge is created, errors

are eliminated, and technologies improve. When pressure is high and capacity is low, collapse occurs. When pressure is low and capacity is high, stagnation follows. When both are low, decay occurs.

5 Reduced-Form Model

This section formalises the interaction between realised problem pressure and problem-solving capacity. The model expresses the core mechanism in reduced form, without microfoundations. It shows how explanatory knowledge is created when societies face solvable problems and possess the capacity to search and survive error. Technology applies that knowledge, output rises, and the attainable frontier moves.

5.1 Actual Output and Frontier

Let Y_t denote actual output or capability at time t : what a society is currently able to produce given its existing knowledge, technology and organisation.

Let $\bar{Y}_t$ denote the attainable frontier at time t : what could be produced if current explanatory knowledge were fully and optimally applied, without creating new knowledge. This distinction is essential. Application of existing knowledge moves Y_t toward $\bar{Y}_t$, while innovation shifts the frontier itself:

$$\bar{Y}_{t+1} = \bar{Y}_t + \delta A_t$$

where $\delta > 0$ measures how efficiently applied knowledge raises the frontier.

5.2 Realised Problem Pressure

The gap $\bar{Y}_t - Y_t$ defines the objective scope for improvement. However, problems must be recognised as solvable to induce search. Define:

$$P_t = \phi_t(\bar{Y}_t - Y_t)$$

where $\phi_t \in [0, 1]$ measures the share of the gap that is perceived, and actionable. For most of human history $\phi_t \approx 0$: problems were interpreted as fate or the will of gods (Deutsch, 2011; Popper, 1959), and not as opportunities for explanation or improvement. Only when societies adopt the epistemic norm that problems are solvable does realised problem pressure increase.

5.3 Knowledge Creation

Let K_t denote the stock of explanatory knowledge at time t . Following Popper (1959), explanatory knowledge grows through conjectures and refutations: errors are eliminated, and deeper theories replace shallower ones.

The rate of knowledge creation is assumed proportional to realised problem pressure and problem-solving capacity:

$$\dot{K}_t = \alpha P_t C_t = \alpha \phi_t (\bar{Y}_t - Y_t) C_t$$

where $\alpha > 0$ is a productivity parameter. Knowledge is created only when societies face recognised problems and possess the capacity to search.

5.4 From Knowledge to Technology

Let A_t denote technological capability. Technology is applied knowledge: implementation of explanatory theories into physical or institutional form (Mokyr, 2002). We assume:

$$\dot{A}_t = \beta \dot{K}_t = \alpha \beta \phi_t (\bar{Y}_t - Y_t) C_t$$

where $\beta > 0$ measures how efficiently explanatory knowledge becomes workable technique.

5.5 Output Growth

Output grows in proportion to technological capability:

$$\dot{Y}_t = \gamma A_t$$

Combining gives the reduced-form growth equation:

$$\dot{Y}_t = \gamma \alpha \beta \phi_t (\bar{Y}_t - Y_t) C_t$$

Growth requires:

- a perceived gap,
- the capacity to solve,
- and the application of explanatory knowledge.

5.6 Frontier Dynamics

Innovation shifts the frontier upward:

$$\bar{Y}_{t+1} = \bar{Y}_t + \delta A_t$$

so that even if current problems are solved, new problems emerge at a higher level of capability. This explains the historical observation that growth is unbounded: solving problems creates new problems.

5.7 Comparative Statics

Two comparative statics follow directly from the growth equation:

$$\dot{Y}_t = \gamma\alpha\beta\phi_t(\bar{Y}_t - Y_t)C_t$$

First, holding capacity constant:

$$\frac{\partial \dot{Y}_t}{\partial P_t} = \gamma\alpha\beta C_t > 0$$

Growth increases with realised problem pressure. For a given capacity, stronger recognition of solvable problems accelerates innovation.

Second, holding problem pressure constant:

$$\frac{\partial \dot{Y}_t}{\partial C_t} = \gamma\alpha\beta P_t > 0$$

Growth increases with capacity. For a given pressure, societies that possess the ability to search and survive error grow faster.

A third comparative static follows with respect to the realisation coefficient:

$$\frac{\partial \dot{Y}_t}{\partial \phi_t} = \gamma\alpha\beta(\bar{Y}_t - Y_t)C_t > 0$$

Growth increases with the extent to which problems are perceived as solvable. Even when the objective gap is large, growth may be negligible if ϕ_t is close to zero. This explains why humans faced abundant problems for hundreds of thousands of years without sustained growth: $\phi_t \approx 0$ for most of history.

A final comparative static concerns changes in the frontier. Differentiating with respect

to $\bar{Y}_t$:

$$\frac{\partial \dot{Y}_t}{\partial \bar{Y}_t} = \gamma \alpha \beta \phi_t C_t > 0$$

As explanatory knowledge shifts the frontier upward, new pressure is generated, inducing further search. This mechanism produces unbounded growth: solving problems creates new problems.

6 Feedback and Frontier Development

The reduced-form model describes how realised problem pressure and problem-solving capacity generate explanatory knowledge and growth at a point in time. This section considers the dynamic feedbacks that emerge as knowledge accumulates, technology advances and the frontier moves. The central idea is that growth is not a one-shot event but a self-reinforcing process in which solving problems creates new problems at higher levels of capability.

6.1 Innovation and the Moving Frontier

Recall that actual output Y_t and the attainable frontier $\bar{Y}_t$ are distinct. At any time t , the frontier captures what could be achieved if existing explanatory knowledge were fully applied. Actual output is typically below this level. Application of existing knowledge moves Y_t towards $\bar{Y}_t$, reducing the gap. Innovation, by contrast, shifts the frontier itself:

$$\bar{Y}_{t+1} = \bar{Y}_t + \delta A_t$$

where A_t is technological capability and $\delta > 0$ measures how effectively new technologies raise the attainable frontier.

As explanatory knowledge K_t increases, technology improves and the frontier shifts upward. This implies that even if Y_t catches up to $\bar{Y}_t$ at a given time, the frontier will move again as knowledge accumulates. The economy never fully “arrives”. New problems appear at higher levels of complexity and ambition. Growth at the frontier is therefore an inherently open-ended process.

6.2 Realised Pressure as a Feedback Variable

Realised problem pressure is defined as:

$$P_t = \phi_t(\bar{Y}_t - Y_t)$$

The gap $(\bar{Y}_t - Y_t)$ is influenced by both application and innovation. When existing knowledge is applied effectively, Y_t rises and the gap narrows. When new knowledge is created, $\bar{Y}_t$ rises and the gap may widen again. The net effect on problem pressure depends on the relative speed of application and innovation.

The coefficient ϕ_t captures the epistemic and cultural component: the extent to which the gap is perceived as a solvable problem. As explanatory knowledge grows, societies may come to see more domains as amenable to improvement. This suggests that ϕ_t can itself be influenced by K_t :

$$\phi_{t+1} = g(\phi_t, K_t)$$

with g increasing in K_t when new explanations spread the norm that problems are solvable. The Enlightenment can be interpreted as a discrete increase in ϕ_t : a shift in epistemology that turned many previously accepted conditions into recognised, solvable problems.

6.3 Capacity and Growth of Capacity

Problem-solving capacity C_t is also endogenous. As output rises, societies can invest more in education, infrastructure, and institutions. This raises human capital, improves resilience and may strengthen inclusive institutions. We can express this feedback as:

$$C_{t+1} = h(C_t, Y_t)$$

where h is increasing in Y_t when additional resources are used to build capacity. In this case, growth reinforces future growth by increasing the ability to search and survive error.

However, rising output can also reduce capacity if stability leads to complacency. Rich societies may prioritise rent protection over experimentation, centralisation over openness, and risk aversion over exploration. In such cases h may be non-monotonic: beyond some level, higher income reduces C_t by weakening variation and resilience. This provides a mechanism for stagnation at high levels of income.

6.4 Virtuous and Vicious Cycles

Combining these feedbacks, we can distinguish several dynamic patterns:

1. **Virtuous cycle of frontier growth.** When ϕ_t is high and increasing, and C_t rises with income, explanatory knowledge accumulates, the frontier shifts upward, and realised problem pressure remains positive. Societies continue to identify new solvable problems and possess the capacity to address them. This corresponds to persistent frontier growth.

2. **Stagnation trap.** When ϕ_t falls and C_t is constant or declining, societies stop treating gaps as solvable problems. Even if objective gaps remain large, realised pressure P_t becomes small. Knowledge creation slows, and growth converges to zero. Rich but complacent societies fit this pattern.
3. **Collapse trap.** When C_t collapses due to shock, war or institutional breakdown, societies lose the capacity to search and survive error. Even if realised pressure is high, $\dot{K}_t \approx 0$ because $C_t \approx 0$. Output falls, the frontier ceases to move, and previously accumulated knowledge may be lost.
4. **Low-level trap.** When both ϕ_t and C_t remain low, societies neither perceive solvable problems nor build capacity. Growth remains close to zero, and the frontier is largely exogenous.

These regimes are not fixed. Societies can move between them when shocks alter ϕ_t or C_t . For example, a political revolution may increase ϕ_t by promoting a culture of criticism, while a war may destroy C_t by eliminating population and institutions.

6.5 Timing of the Break in Growth

The feedback perspective helps explain why sustained growth is historically recent. For most of human history, both ϕ_t and C_t were low. Problems were interpreted as fate, and capacity to search was limited. Explanatory knowledge existed but grew slowly. The Industrial Enlightenment can be interpreted as a joint shift in ϕ_t and C_t : epistemic norms changed, and institutions emerged that allowed more people to participate in search, finance experiments, and survive failure.

Once these feedbacks became positive, growth reinforced itself. Higher output funded more education, research and infrastructure, raising C_t . New explanations expanded the set of perceived solvable problems, raising ϕ_t . The frontier moved continuously, and the modern era of unbounded growth began.

6.6 Path Dependence

Because both ϕ_t and C_t respond to past levels of knowledge and income, the system is path dependent. Societies that experience early gains in explanatory knowledge and capacity may enter virtuous cycles. Societies that experience early collapses may lose knowledge and institutions, making it costly to rebuild capacity. Small differences in initial conditions or shocks can therefore lead to very different long-run outcomes.

This path dependence is consistent with the historical divergence between regions. Some places transitioned into sustained frontier growth; others stagnated or declined. The Problem-Driven Growth Theory interprets these divergences as the result of different trajectories in realised problem pressure and capacity, mediated by feedbacks between knowledge, technology, income and institutions.

The next section uses specific historical cases to illustrate these dynamics.

7 Historical Cases

This section illustrates the model using four historical cases. Each case is interpreted through realised problem pressure P_t , problem-solving capacity C_t , and outcomes. The aim is not to provide full histories, but to show how the mechanism explains long-run patterns.

7.1 Silicon Valley: Frontier Growth

Silicon Valley represents the canonical case of frontier growth. During the Cold War, the space race and digital revolution, realised problem pressure was high. Defence, computing and communication were interpreted as solvable problems. The epistemic norm that technology could solve national and commercial challenges was strong, indicating ϕ_t close to one.

Capacity was also high. Research universities, venture capital, immigration, intellectual freedom and decentralised organisational forms enabled variation, experimentation and survival of error. Failures were tolerated, financial capital was abundant, and successful practices diffused. This created a persistent regime of high P_t and high C_t :

$$P_t > 0, \quad C_t > 0 \quad \Rightarrow \quad \dot{K}_t > 0$$

Explanatory knowledge was created, technologies were implemented, output rose and the frontier shifted repeatedly. New problems emerged: new markets, new scientific challenges, new technological opportunities. Silicon Valley remains a central example of virtuous feedback between pressure, capacity and frontier movement.

7.2 Industrial Manchester: The First Break

Industrial Manchester provides a historical case of frontier growth at the very beginning of the modern era. In the late eighteenth and early nineteenth centuries, the town faced real constraints. Labour was relatively expensive compared to energy, and competition in

textiles was intense. These conditions were not simply endured; they were treated as engineering and organisational challenges. The Industrial Enlightenment had already begun to change the way people thought: nature could be explained, and explanations could be turned into machines. In other words, basic physical and mechanical problems were interpreted as solvable.

Capacity was also emerging. Institutions allowed entry, investment and experimentation. Entrepreneurs could start firms, investors could back them, and inventors could commercialise their ideas. Engineers, mechanics and artisans accumulated practical and theoretical skills. Urban growth and capital accumulation increased resilience. Manchester had enough redundancy and wealth to survive failed experiments and still continue searching. In this setting, strong problem pressure met growing capacity. Steam engines, spinning machinery and iron production techniques were developed and improved. Each success created new problems at a higher level: how to move goods faster, how to finance larger operations, how to manage crowded cities, how to handle sanitation. The result was the first sustained break from the long historical pattern of near zero growth. As in Silicon Valley, high pressure combined with high capacity:

$$P_t > 0, \quad C_t > 0 \quad \Rightarrow \quad \dot{K}_t > 0$$

Steam power, spinning machinery and iron production increased technological capability. The frontier shifted. New problems emerged at higher levels: coordination, transport, finance, urban sanitation. The feedback mechanism began and sustained the first era of modern growth.

7.3 Merv and Baghdad: Destruction of Capacity

Merv and Baghdad illustrate collapse. Both regions possessed explanatory knowledge, technologies, trade networks and economic output. Realised pressure existed. However, resilience was low. When the Mongol invasions destroyed population and institutions, capacity collapsed:

$$C_t \rightarrow 0$$

Even if problems were recognised as solvable, search was impossible. Knowledge and infrastructure disappeared. Trade routes broke, irrigation systems failed and scholarly communities were killed or dispersed. In the model:

$$P_t > 0, \quad C_t \approx 0 \quad \Rightarrow \quad \dot{K}_t \approx 0$$

Collapse was rapid. Rebuilding was slow or impossible. These cases show the asymmetry

between creation and destruction: it takes generations to build capacity but days to eliminate it. Collapse occurs when pressure is high but resilience is insufficient.

7.4 The Dark Ages: Low-Level Trap

After the collapse of the Western Roman Empire, Europe entered a low-level trap. Capacity fell as institutions, infrastructure and literacy declined. At the same time, realised pressure was low. Many problems were interpreted as fate or divine order rather than solvable challenges. The coefficient ϕ_t was small. In this regime:

$$P_t \approx 0, \quad C_t \approx 0 \quad \Rightarrow \quad \dot{K}_t \approx 0$$

Problems existed, but few were interpreted as actionable. Knowledge decayed. Output stagnated. Only later, with the gradual re-emergence of explanatory epistemology, trade and institutional capacity, did Europe begin to escape the trap.

7.5 Summary

The four cases align with the regimes of the model:

Case	Realised Pressure P_t	Capacity C_t	Outcome
Silicon Valley	High	High	Frontier Growth
Manchester	High	High	Frontier Growth
Merv / Baghdad	High	Lost	Collapse
Dark Ages	Low	Low	Low-Level Trap

These cases illustrate that explanatory knowledge grows when societies face recognised, solvable problems and possess the capacity to search, experiment and survive error. When either element is absent, growth slows, stops or reverses. The next section considers implications for the frontier.

8 Predictions for Frontier Economies

The Problem-Driven Growth Theory offers several predictions for economies at or near the technological frontier. These predictions follow from the dynamics of realised problem pressure, problem-solving capacity and feedback between output, knowledge and the frontier.

8.1 Prediction 1: Frontier Growth is Unbounded

If explanatory knowledge continues to grow, the frontier will move without bound. This follows from the feedback equation:

$$\bar{Y}_{t+1} = \bar{Y}_t + \delta A_t$$

Solving problems at one level creates new problems at a higher level. The process is open-ended. There is no theoretical limit to frontier growth except the laws of physics and the current stock of explanatory knowledge. This is consistent with the historical observation that sustained growth is recent but accelerating.

8.2 Prediction 2: Knowledge, Not Resources, Drives Growth

Material resources do not constrain frontier economies. New knowledge enables substitution, saving and discovery. Scarcity itself generates pressure and induces search. The relevant constraint is explanatory knowledge. This implies that:

Resource-rich but knowledge-poor societies will not grow.

Conversely:

Knowledge-rich but resource-poor societies can grow rapidly.

This is consistent with the experience of small frontier hubs such as Silicon Valley, Tel Aviv and Singapore.

8.3 Prediction 3: Declining Realised Pressure Leads to Stagnation

Frontier societies that solve many problems may experience a decline in realised problem pressure. If ϕ_t falls, search slows even if objective gaps remain. Welfare states, comfort and stability may reduce the perception of solvable problems. The model predicts:

$$\phi_t \downarrow \Rightarrow \dot{K}_t \downarrow$$

This offers an explanation for episodic stagnation in rich societies that is distinct from demographic, monetary or resource-based explanations.

8.4 Prediction 4: Collapse is a Risk When Capacity is Fragile

Capacity C_t includes resilience. If institutions, infrastructure or population are destroyed, capacity may fall to zero. Under such conditions, high pressure produces collapse rather than innovation:

$$P_t > 0, \quad C_t \rightarrow 0 \quad \Rightarrow \quad \dot{K}_t \approx 0$$

This prediction applies to fragile frontier states where single shocks can eliminate knowledge and institutions. Collapse is easier than creation.

8.5 Prediction 5: Frontier Growth Requires a Culture of Solvability

The coefficient ϕ_t captures epistemic norms: the belief that problems are solvable. Frontier growth requires:

$$\phi_t \approx 1$$

Popper's logic of error-correction and Deutsch's notion of explanatory knowledge imply that societies must tolerate criticism, experimentation and revision. If problems are interpreted as fate, divine will or tradition, realised pressure will remain low even when conditions are poor. This is consistent with the long period of flat growth in early human history.

8.6 Prediction 6: Policies That Increase ϕ_t and C_t Have Outsized Effects

The theory implies that the most effective policies at the frontier are those that increase realised problem pressure or capacity. Policies that raise ϕ_t include:

- deep scientific education,
- open criticism,
- freedom of speech,
- transparency,
- protection of dissent.

Policies that raise C_t include:

- investment in research,
- decentralised decision-making,

- immigration of talent,
- institutions that allow entry,
- reserves and redundancy.

These are not traditional macroeconomic levers. They target the epistemic and institutional components of capacity.

8.7 Prediction 7: Frontier Leaders Will Be Those With High ϕ_t and High C_t

Societies that sustain high realised pressure and high capacity will remain at the frontier. These conditions are most likely where:

- problems are recognised and believed solvable,
- institutions support experimentation and resilience,
- knowledge grows through error correction.

The model predicts that frontier leadership will shift toward societies that combine:

$$\phi_t \approx 1, \quad C_t \approx 1$$

and away from those where either term declines.

8.8 Prediction 8: Convergence Requires Epistemology, Not Only Capital

Developing countries catch up when they adopt existing technologies. The speed of convergence depends on human capital and institutions (Nelson and Phelps, 1966; Benhabib and Spiegel, 1994). The present theory extends this by adding ϕ_t : the belief in solvability. Convergence requires:

$$\phi_t > 0, \quad C_t > 0$$

Societies without epistemic norms of solvability may remain in low-level traps, even with access to capital and ideas.

8.9 Prediction 9: Policy That Ignores Knowledge Will Misdiagnose the Frontier

Mainstream growth theory often treats knowledge as a residual. The present theory predicts that this is a mistake. Frontier growth is driven by explanatory knowledge. Institutions, infrastructure and resources matter only insofar as they increase ϕ_t and C_t . Policies that ignore epistemology, resilience and error-correction will underestimate the determinants of frontier growth.

8.10 Summary

The Problem-Driven Growth Theory predicts:

1. Frontier growth is unbounded.
2. Knowledge, not resources, drives long-run growth.
3. Declining realised pressure leads to stagnation.
4. Collapse occurs when capacity is fragile.
5. Frontier growth requires a culture of solvability.
6. Policies that increase ϕ_t and C_t are most effective.
7. Frontier leadership will shift toward societies with high ϕ_t and high C_t .
8. Convergence requires epistemology as well as capital.
9. Policies that ignore knowledge misdiagnose the frontier.

These predictions arise directly from the reduced-form model:

$$\dot{K}_t = \alpha \phi_t (\bar{Y}_t - Y_t) C_t$$

Growth at the frontier occurs when societies face recognised, solvable problems and possess the capacity to search, experiment and survive error.

9 Policy Implications

The theory implies that the most effective policies at the frontier are those that increase realised problem pressure ϕ_t or problem-solving capacity C_t . Conventional macroeconomic levers that target investment, interest rates or physical capital are not the primary drivers

of frontier growth. Growth is driven by explanatory knowledge, and policies should focus on its creation, diffusion and application.

9.1 Policy Lever 1: Increase Realised Problem Pressure

Problem pressure exists when societies recognise solvable problems. The coefficient ϕ_t measures this realisation. Policies that increase ϕ_t do not create problems, but increase the perception that problems can be solved. These include:

- scientific education that emphasises explanation rather than memorisation,
- public communication of unsolved challenges,
- open access to data and transparency,
- incentives for innovation prizes and competitions,
- decentralisation that exposes local problems.

The aim is not dissatisfaction or anxiety, but epistemic optimism. Problems should be visible and interpreted as solvable. Historically, the Enlightenment increased ϕ_t by promoting the idea that nature, society and technology could be understood and improved.

9.2 Policy Lever 2: Increase Problem-Solving Capacity

Capacity C_t enables societies to search, experiment and survive error. Policies that increase capacity include:

- investment in research, laboratories and scientific institutions,
- protection of freedom of speech and criticism,
- immigration of talent and scientific labour mobility,
- funding for early-stage experimentation,
- decentralisation of decision-making.

Resilience is a component of capacity. Redundancy, buffers and reserves allow societies to absorb failures. Policies that reduce fragility increase C_t .

9.3 Policy Lever 3: Protect Variation and Entry

Innovation requires variation. Policies that restrict entry reduce variation and therefore reduce innovation. This principle applies to:

- labour markets,
- firm formation,
- finance,
- research agendas.

Barriers to entry, professional cartels and bureaucratic licensing reduce experimentation and slow error-correction. Policies should favour openness, competition and experimentation.

9.4 Policy Lever 4: Strengthen Institutions of Error-Correction

Following Popper, knowledge grows when errors are eliminated. Policies that strengthen error-correction include:

- independent scientific institutions,
- academic freedom,
- peer review,
- judiciary independence,
- transparent auditing.

Error must not be punished. Societies that suppress dissent reduce ϕ_t and C_t simultaneously. Error-correction is the engine of progress.

9.5 Policy Lever 5: Invest in Explanatory Knowledge

Explanatory knowledge generalises across contexts. It has high spillovers and long-run value. Policies should prioritise:

- fundamental science,
- mathematics,
- theoretical physics,

- computer science,
- epistemology and scientific reasoning.

These fields produce deep explanations that enable the solution of unfamiliar problems. They increase $\bar{Y}_t$ by shifting the frontier itself.

9.6 Policy Lever 6: Increase Resilience Against Collapse

Collapse occurs when capacity falls to zero. The model predicts that collapse is easier than creation. Policies should build resilience through:

- diversification of supply chains,
- infrastructure maintenance,
- reserves of energy, food and capital,
- military capability,
- social trust and cohesion.

These policies are not growth-maximising in the short run. They prevent $\dot{K}_t$ from falling to zero in the long run.

9.7 Policy Lever 7: Focus on the Frontier

Policies should be evaluated at the frontier. Once basic needs are met, marginal improvements in physical capital have diminishing returns. Growth depends on explanatory knowledge. Policies should focus on:

- new knowledge,
- new explanations,
- new technologies.

Policies that reallocate resources from rent-preservation to innovation increase both ϕ_t and C_t .

9.8 Summary

The theory implies that frontier growth is not driven by resources, demographics or capital accumulation, but by explanatory knowledge. The most effective policies increase the extent to which problems are perceived as solvable and the capacity to solve them. The relevant policy instruments are epistemic and institutional rather than purely economic. The goal is to sustain the norms of criticism, experimentation and resilience that allow knowledge to grow.

10 Conclusion

This paper has proposed a reduced-form theory of frontier growth. The central mechanism is that explanatory knowledge is created when societies face recognised, solvable problems and possess the capacity to search, experiment and survive error. Realised problem pressure and problem-solving capacity are necessary and jointly sufficient for knowledge creation. The main equation is:

$$\dot{K}_t = \alpha \phi_t (\bar{Y}_t - Y_t) C_t$$

Knowledge grows through error-correction (Popper, 1959). Technologies apply knowledge, output rises and the frontier moves:

$$\bar{Y}_{t+1} = \bar{Y}_t + \delta A_t$$

Solving problems creates new problems at higher levels of capability. Growth at the frontier is therefore open-ended. Collapse occurs when capacity fails. Stagnation occurs when realised problem pressure falls. Low-level traps persist when both terms remain small. These outcomes are unified by a single mechanism.

The theory explains historical patterns. Frontier growth in Manchester and Silicon Valley occurred when P_t and C_t were high. Collapse in Merv and Baghdad occurred when C_t fell to zero. The Dark Ages persisted as a low-level trap when few problems were interpreted as solvable and capacity was weak. These dynamics emerge without appealing to exogenous technology shocks, geographic determinism or resource constraints.

The policy implications differ from conventional macroeconomics. The most effective levers increase the culture of solvability (ϕ_t) or capacity (C_t). Policies that support science, criticism, open entry, resilience and decentralisation are more important than policies that simply increase investment or capital. Epistemology, not physical resources, is central at the frontier.

The main limitation of the model is its reduced-form nature. Microfoundations are not

specified. The behavioural and institutional processes that generate ϕ_t and C_t merit further research. Future work should formalise these components, develop empirical measures and test the predictions.

The conclusion is optimistic. Progress is driven by explanatory knowledge. There is no theoretical limit to growth except the laws of physics and the current stock of knowledge. If societies maintain the norms that enable criticism and error-correction, the frontier will continue to move. The future depends not on what exists, but on what problems we choose to solve.

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